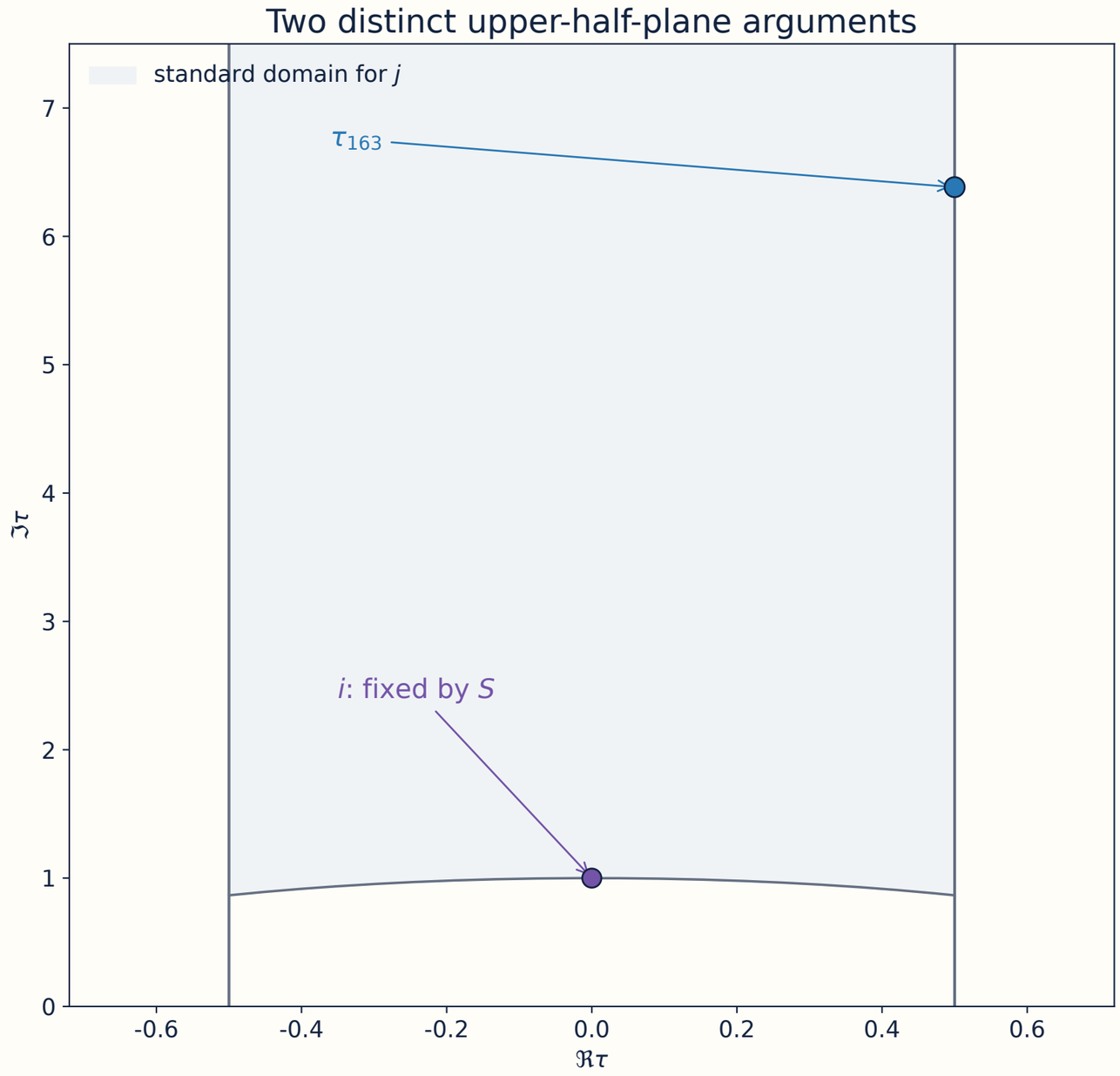
Figure 10. Two CM arguments, two roles: inversion fixed point versus Hermite’s small nome



Golden geometry = gold | Fourier / theta = teal | modular map = violet | CM / Hermite = blue

i

$q = e^{-2\pi}$

$j(i) = 1728$

fixed point selects $r(i) = \kappa$

$\tau_{163} = (1 + i\sqrt{163})/2$

$q = -e^{-\pi\sqrt{163}},\ j = -640320^3$

integral j-value and tiny negative nome